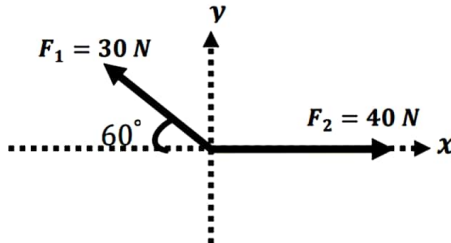




SET B

TRIAL PSPM1 DP014 SESI 2024/2025

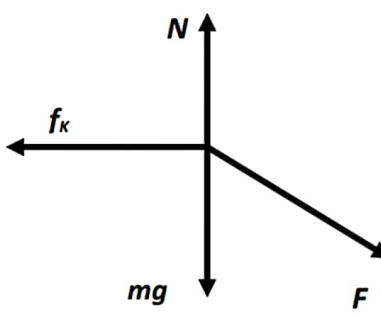
SKF

NO.	SOLUTION & ANSWER												
1 (a)(i)	$Volume = 0.06\text{ m} \times 0.12\text{ m} \times 0.22\text{ m}$ $= 1.58 \times 10^{-3}\text{ m}^3$												
1(a)(ii)	$Density = \frac{mass}{volume}$ $= \frac{8.50\text{ kg}}{1.584 \times 10^{-3}\text{ m}^3}$ $= 5.37 \times 10^3\text{ kg m}^{-3}$												
1(b)	 <table><tr><th>Vector</th><th>x-component</th><th>y-component</th></tr><tr><td>F_1</td><td>$-F_1 \cos 60^\circ$ $= -30 \cos 60^\circ$ $= -15\text{ N}$</td><td>$F_1 \sin 60^\circ$ $= 30 \sin 60^\circ$ $= 25.98\text{ N}$</td></tr><tr><td>F_2</td><td>$F_2 = 40\text{ N}$</td><td>0</td></tr><tr><td>Magnitude, F_R</td><td>$\sum F_x = 25\text{ N}$</td><td>$\sum F_y = 25.98\text{ N}$</td></tr></table> Magnitude, $F_R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$ $= \sqrt{(25)^2 + (25.98)^2}$ $= 36.05\text{ N}$ Direction, θ $\tan \theta = \frac{\sum F_y}{\sum F_x}$ $= \frac{25.98}{25} = 1.0392$ $\theta = 46.10^\circ \text{ above positive x-axis}$	Vector	x-component	y-component	F_1	$-F_1 \cos 60^\circ$ $= -30 \cos 60^\circ$ $= -15\text{ N}$	$F_1 \sin 60^\circ$ $= 30 \sin 60^\circ$ $= 25.98\text{ N}$	F_2	$F_2 = 40\text{ N}$	0	Magnitude, F_R	$\sum F_x = 25\text{ N}$	$\sum F_y = 25.98\text{ N}$
Vector	x-component	y-component											
F_1	$-F_1 \cos 60^\circ$ $= -30 \cos 60^\circ$ $= -15\text{ N}$	$F_1 \sin 60^\circ$ $= 30 \sin 60^\circ$ $= 25.98\text{ N}$											
F_2	$F_2 = 40\text{ N}$	0											
Magnitude, F_R	$\sum F_x = 25\text{ N}$	$\sum F_y = 25.98\text{ N}$											



NO.	SOLUTION & ANSWER
2(a)(i)	$v_{0-2} = \frac{18-0}{2-0} = 9 \text{ m s}^{-1}$ $v_{2-7} = \frac{18-18}{7-2} = 0 \text{ m s}^{-1}$ $v_{7-10} = \frac{0-18}{10-7} = -6 \text{ m s}^{-1}$
2(a)(ii)	
2(b)	$s = ut + \frac{1}{2}at^2$ $30 = (u)(3) + \frac{1}{2}(2)(3)^2$ $u = 7 \text{ m s}^{-1}$
2(c)	$s = ut + \frac{1}{2}at^2$ $= (15)(2) + \frac{1}{2}(-9.81)(2)^2$ $= 10.38 \text{ m}$ above released point
	JUML

NO.	SOLUTION & ANSWER
3(a)(i)	$J = mv - mu$ $= (0.04)(0) - (0.04)(-30.67)$ $= 1.23 \text{ kgms}^{-1} \text{ or } 1.23 \text{ Ns}$
3(a)(ii)	3 / 8 ••• $J = Ft$ $1.23 = F(0.03)$ $F = 41 \text{ N}$
3(b)	$\Sigma p_i = \Sigma p_f$ $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$ $(1500)(75) + 2000 u_2 = 0$ $u_2 = -56.2 \text{ ms}^{-1}$
3(c)	$\Sigma p_{iy} = \Sigma p_{fy}$ $m_1 u_{1y} + m_2 u_{2y} = m_1 v_{1y} + m_2 v_{2y}$ $0 = (500)(20 \sin 30) + 200(-v_2 \sin 6.9)$ $v_2 = 208.1 \text{ ms}^{-1}$

NO.	SOLUTION & ANSWER
4(a)	$\sum F_y = 0$ $T_p \sin 50^\circ - 16(9.81) = 0$ $T_p = 204.9 \text{ N}$ $\sum F_x = 0$ $T_Q - 204.9 \cos 50^\circ = 0$ $T_Q = 131.7 \text{ N}$
4(b)(i)	 <i>Label and direction</i> <i>All correct ----- 3 marks</i> <i>3 correct ----- 2 marks</i> <i>2 correct ----- 1 mark</i> 4 / 8
4(b)(ii)	$\sum F_x = ma$ $64 \sin 55^\circ - f_k = 3.7(0.8)$ $f_k = 49.47 \text{ N}$
4(b)(iii)	$\sum F_y = 0$ $N - 3.7(9.81) - 64 \cos 55^\circ = 0,$ $N = 73.01 \text{ N}$ $f_k = \mu N$ $\mu = \frac{f_k}{N} = \frac{49.47}{73.01} = 0.68$
	JUN

NO.	SOLUTION & ANSWER
5(a)(i)	$T = \frac{\text{time taken}}{\text{no. of revolution}}$ $0.25 = \frac{\text{time taken}}{75}$ $\text{time taken} = 18.75 \text{ s}$
5(a)(ii)	$\frac{\theta}{t} = \frac{2\pi}{T}$ $\frac{\theta}{2(60)} = \frac{2\pi}{0.25}$ $\theta = 3015.93 \text{ rad OR } 480 \text{ rev}$
5(a)(iii)	$v = \frac{2\pi r}{T}$ $= \frac{2\pi(0.3)}{0.25}$ $= 7.54 \text{ m s}^{-1}$
5(b)(i)	$22.4 \text{ rpm} = \frac{22.4(2\pi)}{60} = 2.35 \text{ rad s}^{-1}$
5(b)(ii)	$\omega = \frac{\theta}{t}$ $2.35 = \frac{\theta}{150}$ $\theta = 352.5 \text{ rad}$
5(b)(iii)	$f_k = F_c \rightarrow \mu_k mg = mv\omega$ $\mu_k(9.81) = (3.5)(2.35)$ $\mu_k = 0.84$
5(b)(iv)	$v = r\omega$ $3.5 = r(2.35)$ $r = 1.48 \text{ m}$
	JUI

NO.	SOLUTION & ANSWER
6(a)(i)	$u = r\omega_0$ $17.0 = 0.48(\omega_0)$ $\omega_0 = 35.4 \text{ rads}^{-1}$ $a = r\alpha$ $2.00 = 0.48\alpha$ $\alpha = 4.17 \text{ rads}^{-2}$ $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$ $= (35.4)(5.00) + \frac{1}{2}(4.17)(5.00)^2$ $= 229 \text{ rad}$ $\text{Number of revolution} = \frac{\theta}{2\pi} = \frac{229}{2\pi} = 36.4 \text{ revolutions}$
6(a)(ii)	$\omega = \omega_0 + \alpha t$ $= 35.4 + (4.17)(5.00)$ $= 56.3 \text{ rads}^{-1}$
6(b)	$\omega^2 = \omega_0^2 + 2\alpha\theta$ $0 = (35.4)^2 + 2(-4.17)\theta$ $\theta = 150.26 \text{ rad}$ $\text{Number of revolution} = \frac{\theta}{2\pi} = \frac{150.26}{2\pi} = 23.9 \text{ revolutions}$
6(c)	Considering the center of concentric wheels, $\tau_{\text{nett}} = -\tau_{35N} + \tau_{28N} - \tau_{18N}$ $\tau_{\text{nett}} = -(35)(0.12)\sin 135 + (28)(0.24)\sin 90 - (18)(0.24)\sin 90$ $\tau_{\text{nett}} = -0.57 \text{ Nm}$ Clockwise

NO.	SOLUTION & ANSWER
7(a)(i)	$\frac{dT}{x} = \frac{303.15 - 373.15}{0.1}$ $\frac{dT}{x} = -700 K m^{-1}$
7(a)(ii)	$\frac{dQ}{dt} = -kA \frac{dT}{x}$ $\frac{dQ}{dt} = -(397)(2 \times 10^{-4})(-700)$ $\frac{dQ}{dt} = 55.58 W$
7(a)(iii)	$\frac{dQ}{dt} = 55.58 W$ $\frac{dQ}{2 \times 60} = 55.58$ $dQ = 6.67 \times 10^3 J$
7(b)(i)	$v_{rms} = \sqrt{\frac{3RT}{M}}$ $v_{rms} = \sqrt{\frac{3(8.31)(298.15)}{0.032}}$ $v_{rms} = 481.96 m s^{-1}$
7(b)(ii)	$v_{rms} = \sqrt{\frac{3RT}{M}}$ $600 = \sqrt{\frac{3(8.31)T}{0.032}}$ $T = 462.09 K$
7(c)(i)	$Q = \Delta U + W$ $350 = 150 + W$ $W = 200 J$



7(c)(i)	$Q = \Delta U + W$ $350 = 150 + W$ $W = 200J$
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7(c)(ii)	Volume increase
7(d)(i)	PQ = isobaric QR = isochoric/isovolumetric $Q = \Delta U + W$ $12000 = \Delta U + 6000$ $\Delta U = 6000J$
7(d)(ii)	W= 0 $Q = \Delta U + W$ $Q = 4000 + 0$ $Q = -4000J$
	JUN

